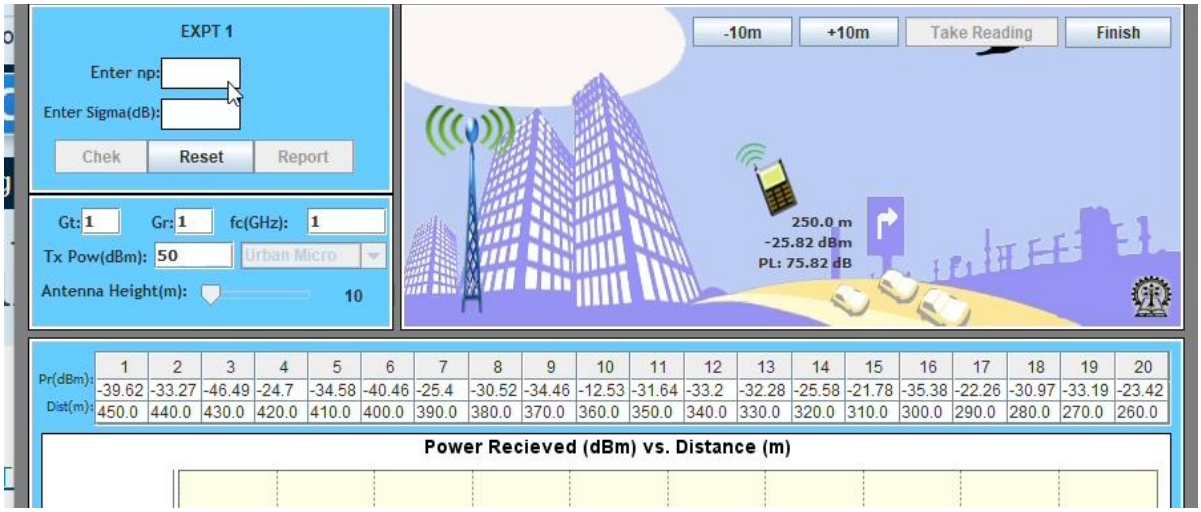
Steps to calculate **n** and **sigma** using calculation

1. STEP 1: To find power at different distance using below formula

$$P_r(d_i) = P_r(d_0) - 10n \log_{10}\left(\frac{d_i}{d_0}\right)$$

Where, i = 1 to 20 (for our experiment)



	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
Pr(dBm):	-39.62	-33.27	-46.49	-24.7	-34.58	-40.46	-25.4	-30.52	-34.46	-12.53	-31.64	-33.2	-32.28	-25.58	-21.78	-35.38	-22.26	-30.97	-33.19	-23.42
Dist(m):	450.0	440.0	430.0	420.0	410.0	400.0	390.0	380.0	370.0	360.0	350.0	340.0	330.0	320.0	310.0	300.0	290.0	280.0	270.0	260.0

d_i = distance

$P_r(d_i) =$ power at distance di

$P_r(d_0) = 0$ dBm at $d_0 = 100$ m (assumption for our experiment)

So we will have $P_r(d_1)$ to $P_r(d_{20})$ in terms of n

$$P_r(d_1) = 0 - 10n \log_{10}\left(\frac{d_1}{100}\right) = -10n\, log_{10}\left(\frac{450}{100}\right) = -6.53212n$$

$$P_r(d_2) = 0 - 10n\, log_{10}\left(\frac{d_2}{100}\right) = -6.43453n$$

$$P_r(d_3) = 0 - 10n\, log_{10}\left(\frac{d_3}{100}\right) = -6.33468n$$

$$P_r(d_4) = 0 - 10n\, log_{10}\left(\frac{d_4}{100}\right) = -6.23249n$$

$$P_r(d_5) = 0 - 10n\, log_{10}\left(\frac{d_5}{100}\right) = -6.12784n$$

$$P_r(d_6) = 0 - 10n\, log_{10}\left(\frac{d_6}{100}\right) = -6.02060n$$

Here d_1 to d_6 is from experiment

2. Now calculate Minimum mean square error (MMSE) and n
   MMSE is J(n)

$$J(n) = \sum_{i=1}^{k} \left(\left(P_r - \widehat{P_r} \right) \right)^2$$

Where,
P_r is the computed value.
$\widehat{P_r}$ is the calculated value.

computed value is from experiment
calculated is from above step 1.

Here $J(n)$ is calculated

$$J(n) = (-39.62 + 6.53213n)^2$$
$$(1569.744 - 258.79n + 42.667n^2)$$
$$+(-33.27 + 6.43453n)^2$$
$$(1106.893 - 214.0592n + 41.396n^2)$$
$$+(-46.49 + 6.33468n)^2$$
$$(2161.3201 - 294.2817n + 40.1322n^2)$$
$$+(-24.70 + 6.23249n)^2$$
$$(610.09 - 153.94n + 38.8439n^2)$$
$$+(-34.58 + 6.12784n)^2$$
$$(1195.7764 - 211.900n + 37.550n^2)$$
$$+(-40.46 + 6.02060n)^2$$
$$(1637.0116 - 243.5934n + 36.2476n^2)$$

Use the below formula on each square

$$(a + bn)^2 = a^2 + 2abn + b^2n^2$$

so we get finally $J(n)$ as

$$J(n) = 236.8367n^2 - 1{,}376.5643n + 8{,}280.8351$$

Differentiate with respect to n:

$$\frac{dJ(n)}{dn} = 2(236.8367)n - 1{,}376.5643$$

$$\frac{dJ(n)}{dn} = 473.6734n - 1{,}376.5643$$

$$\boxed{\frac{dJ(n)}{dn} = 0}$$

$$473.6734n - 1{,}376.5643 = 0$$

$$n = \frac{1{,}376.5643}{473.6734}$$

$$n = 2.9061$$

Now $J(n)$ is

$$J(n) = 236.8367n^2 - 1{,}376.5643n + 8{,}280.8351$$

$$J(5.59245) = 6{,}280.5863$$

3. Now find sigma using below formula

$$\sigma^2 = \frac{J(n)}{N}$$

$$N =$$

$$\sigma^2 = \frac{6{,}280.5863}{6}$$

$$\sigma^2 = 1{,}046.7643 \text{ dB}^2$$

$$\sigma = \sqrt{1{,}046.7643}$$

$$\sigma = 32.35\text{dB}$$

(Note : - In exam, only 6 readings are enough for path loss fading. so, we will have i = 1 to 6. N = 6. Rest Entire procedure remains the same. This all above calculation we have to do on paper with calculator)

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Frequency reuse

Just use this formula to verify value of N

$$N = i^2 + ij + j^2$$

for us i=2 and j=1 . therefore $N = 2^2 + 2X1 + 1^2$=7